Hertz | LLM Algorithm Engineer • Compression • Multimodal

LLM • PyTorch • DeepSpeed • RAG • Agent • Compression

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*Master's student in Computer Science at Tsinghua University, specializing in Large Language Models with deep focus on **model compression** and **efficient fine-tuning**. First-author NeurIPS paper, Kaggle LLM gold medal (Top 1%), and 500+ GitHub stars across open-source projects. End-to-end experience from data construction → training → inference deployment.*

Skills

Programming Python (expert) • C++ / CUDA • Bash • TypeScript

LLM Stack LLM | SFT | RLHF/DPO | LoRA/QLoRA | Distillation | Quantization | RAG | Agent | Multimodal

Frameworks PyTorch • HuggingFace • DeepSpeed / FSDP • vLLM / TensorRT-LLM • LangChain / LlamaIndex

Engineering Linux • Git • Docker / K8s • Ray • Slurm • W&B / MLflow • AWS / Aliyun

Data Stores PostgreSQL • Redis • Elasticsearch • Milvus / Qdrant • Kafka • Spark

Natural Languages Chinese (native) • English (CET-6, can read & write papers and documentation)

Education

Tsinghua University • Department of Computer Science Sep 2023 – Jun 2026 (expected)

M.S. in Computer Science Top 5% in cohort; advisor's focus: large language models

- **Core courses**: Deep Learning, NLP, Reinforcement Learning, Distributed Systems, Advanced Algorithms
- **Honors**: National Scholarship 2023; Excellence Scholarship (1st class) 2024; Outstanding Graduate Student

Tsinghua University • Department of Computer Science Sep 2019 – Jun 2023

B.S. in Computer Science GPA 3.85/4.00; exchange at ETH Zürich, Spring 2022 (grade 5.5/6.0)

Internships

ByteDance • Doubao LLM Team Jun 2024 – Dec 2024

LLM Algorithm Intern

- Led CoT data construction for code/math scenarios (5B tokens for continued pre-training); HumanEval / MATH / GSM8K averages **+5–8 pp**
- Designed multi-task SFT + DPO recipe (LLaMA-3-8B) over 500K multi-turn dialogues; internal benchmark **+3 pp**, user satisfaction **87%**
- Built ReAct + Tree-of-Thought hybrid reasoning framework; tool-call accuracy **72% → 89%**, complex-task success **+12%**

Tencent • WeChat AI Jun 2023 – Dec 2023

Algorithm Engineering Intern

- Optimised inference (KV-cache reuse + kernel fusion + dynamic batching); P99 **3.2s → 1.8s**, QPS **+65%**
- Implemented INT8 + mixed-precision quantization; **98%** performance retained, memory **−55%**, deployments per machine doubled
- Extended vLLM with continuous batching and preemptive scheduling; throughput **+80%** at 1000+ QPS, P95 **−40%**

Research Experience

Tsinghua NLP Lab • LLM Group Sep 2023 – Present

Research Assistant (1st / 2nd author)

- **RL-driven autonomous decision Agent** (NeurIPS 2026 submission): PPO + hierarchical RL, **+12–18%** task success on ALFWorld / WebShop, **−25%** decision steps; MCTS+RL hybrid yields **+35%** sample efficiency on long-horizon tasks
- **Hierarchical KD for LLM compression** (NeurIPS 2024 CCF-A, published, 15 cites): triple alignment of attention/hidden/output, LLaMA-13B → **3B** retaining **95%**; **+4.2%** over baselines on MMLU/GSM8K

Projects

Self-Correcting RAG • `SelfRAG-Agent` Jan 2024 – May 2024

Enterprise RAG Agent — recall 65% → 85%, in production

- **Stack**: LangChain, LlamaIndex, Qdrant, bge-reranker, FastAPI
- **Approach**: iterative retrieval with ReAct-style reflection + hybrid retrieval (vector + BM25) + cross-encoder rerank
- **Ablation**: iter **+15%** • hybrid **+3%** • rerank **+2%** on MS MARCO / NQ

github.com/hertz/self-rag-agent rag.hertz.dev

Hierarchical-Memory Long Conversation Agent • `HMem-Agent` Jul 2023 – Dec 2023

Working / Short-term / Long-term tri-layer memory inspired by human cognition

- Architecture: Working Memory (last K turns) + Short-term (auto-summary) + Long-term (importance + decay)
- Result: retention **92%** (vs 68%), retrieval **450ms → 150ms**, 100+ turn task success **+28%**

github.com/hertz/hmem-agent

Open-Source

hertz/Agent-RL-Framework ★562 84 Python

RL-driven LLM Agent decision framework (companion to NeurIPS submission)

- PPO / DPO / GRPO across ALFWorld / WebShop / SciWorld; closed 30+ issues, merged 12 external PRs

huggingface/transformers ★135k 27k Python

HuggingFace flagship repo • contributor

- Fixed Trainer.evaluate FSDP OOM (PR #28xxx); added Qwen2 long-context evaluation scripts

vllm-project/vllm ★32k 4.8k Python

High-performance LLM inference engine • contributor

- Optimized PagedAttention scheduling (**+15%** end-to-end throughput); added Baichuan / Yi support

Publications & Patents

Hierarchical Knowledge Distillation for Compact Large Language Models
Hertz, S. Li, W. Wang
NeurIPS 2024 (CCF-A) Paper Code

Self-Reflective Agents: Reinforcement Learning for Long-Horizon Decision Making
Hertz, L. Zhao, W. Chen
NeurIPS 2026 (under review) Preprint Code

Retrieval-Augmented Generation with Iterative Self-Correction
S. Li, **Hertz**, W. Wang
EMNLP 2024 Findings (CCF-B) Paper

Patent • Hierarchical-Distillation LLM Compression Method May 2024
CN202410xxxxx.X • 1st inventor

Patent • Self-Reflective Agent Decision Method Nov 2024
CN202411xxxxx.X • 1st inventor

Honors & Competitions

Kaggle LLM Science Exam • Gold Medal Aug 2023
Top 1% (128/2664); featured in official Kaggle tutorial

Aliyun Tianchi LLM Fine-tuning Challenge • 1st place (national) May 2023
72 teams, prize 50K RMB; few-shot alignment & prompt engineering

Tsinghua National Scholarship / Excellence Scholarship 2023 / 2024
Top 5% in cohort, two consecutive years

© 2025–2026 Hertz • Built with LaTeX

CCPC China Collegiate Programming Contest • National Silver Nov 2022

Northeast Regional 12/380 • Team `ThuTeam`

Mathematical Contest in Modeling (MCM/ICM) • Meritorious Winner Feb 2021

Top 8% globally; topic: epidemic dynamics forecasting

GitHub OSS Influence 2023 – Present

5,000+ Stars across personal repos; flagship `Agent-RL-Framework` 562 stars